

Name

Age / Gender

Contact No.

Address

Pin code

: Mr. SANTOSH VENKITACHALAM

: 50 Year(s)/ Male

: +919920105605

: A 502 MULUND KUMAR MHADA COLONY M...

: 400081

VID No.

PID No.

Referred by

Sample Collected At

: 250067508503480

: P18825553373078

: DR.ABHIJEET JADHAV

: Hcl Avitas Private Limited, 806,siddharth 96 Nehru Place,110019

SUMMARY REPORT

Investigation Outside Reference Range (Abnormal)			
Investigation	Observed Value	Unit	Biological Reference Interval
Absolute Eosinophil Count	1590	cells/cu.mm	20-500
Lymphocytes	19.3	%	20-40
Eosinophils	15.9	%	1-6
Medical Remarks : Eosinophilia			
Glucose Fasting (Plasma-F Hexokinase)	124	mg/dL	Normal: 70-99 Impaired Fasting Glucose(IFG): 100-125 Diabetes mellitus: >= 126 (on more than one occassion) (American diabetes association guidelines 2022)
HbA1c- Glycated Haemoglobin	5.8	%	Non-diabetic: <= 5.6 Pre-diabetic: 5.7-6.4 Diabetic: >= 6.5 Refer interpretation for monitoring ranges.
HDL Cholesterol Direct (Serum Homogeneous Enzymatic colorimetric Assay)	36	mg/dL	> 40
LDL/HDL Ratio (Serum Calculated)	2.2		2.5-3.5
Bilirubin Direct	0.36	mg/dL	0.0-0.3

MEDICAL LABORATORY REPORT



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Pin code	: 400081		

Investigation Outside Reference Range (Abnormal)			
Investigation	Observed Value	Unit	Biological Reference Interval
eGFR	88	ml/min/1.73 sq m	Normal OR high: >= 90 Mild decrease : 60-89 Mild moderate decrease: 45-59 Moderate to severe decrease: 30-44 Severe decrease: 15-29 Kidney failure: < 15
Vitamin D Total - 25 Hydroxy (OH) (Serum Electrochemiluminescence immunoassay (ECLIA))	26.10	ng/mL	Insufficiency: 20-30 Sufficiency: 30-100 Hypervitaminosis: > 100 Deficiency: < 20

Investigation Within Reference Range (Normal)	
Electrolytes, Serum	
Proteins, Serum	
Thyroid Profile - 1	
Iron Studies	
Alkaline Phosphatase, Serum	
BUN, Serum	
Calcium, Serum	
Gamma GT (GGTP)	
LDH (Lactate Dehydrogenase), Serum	
Phosphorus, Serum	
RA (Rheumatoid Arthritis) Factor, Serum	
SGOT (AST)	
SGPT (ALT)	
Uric Acid, Serum	
Vitamin B12 (Cyanocobalamin)	
ESR - Erythrocyte Sedimentation Rate,EDTA	

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CBC Haemogram

Investigation	Observed Value	Unit	Biological Reference Interval
<u>Erythrocytes</u>			
Haemoglobin (Hb)	15.3	gm/dL	14-18
Erythrocyte (RBC) Count	4.92	mill/cu.mm	4.4-6.0
PCV (Packed Cell Volume)	45.2	%	42-52
MCV (Mean Corpuscular Volume)	91.9	fL	82-101
MCH (Mean Corpuscular Hb)	31.2	pg	27-34
MCHC (Mean Corpuscular Hb Conc.)	33.9	gm/dL	31.5-36
RDW (Red Cell Distribution Width)	13.2	%	11.5-14.0
<u>RBC Morphology</u>			
Remark	Normocytic Normochromic		
<u>Leucocytes</u>			
Total Leucocytes Count (TLC)	10,000	cells/cu.mm	4300-10300
Absolute Neutrophil Count	5750	cells/cu.mm	2000-7000
Absolute Lymphocyte Count	1930	cells/cu.mm	1000-3000
Absolute Monocyte Count	710	cells/cu.mm	200-1000
Absolute Eosinophil Count	1590	cells/cu.mm	20-500
Absolute Basophil Count	20	cells/cu.mm	20-100
Neutrophils	57.5	%	40-80
Lymphocytes	19.3	%	20-40
Monocytes	7.1	%	2-10
Eosinophils	15.9	%	1-6
Comments : Eosinophilia			
Basophils	0.2	%	0-2
<u>Platelets</u>			
Platelet count	338	10^3 / μl	140-440
MPV (Mean Platelet Volume)	8.0	fL	7.8-11
PCT (Platelet crit)	0.272	%	0.2-0.5
PDW (Platelet Distribution Width)	16.6	%	9-17

EDTA Whole Blood-Tests done on Automated Five Part Cell Counter. (Haemoglobin by photometric measurement/SLS method, RBC and Platelet count by impedance/Hydrodynamic focusing,WBC and differential by VCS technology/Impedance/Flow cytometry.Rest are calculated parameters).All Abnormal Haemograms are reviewed confirmed microscopically.Differential count is based on approximately 10,000 cells.



Dr. Namrata Singh

Dr. NAMRATA SINGH
 DNB (Pathology)
 Consultant Pathologist
 Reg No.2016083430

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METROPOLIS HEALTHCARE LTD,UNIT NO409-416,4TH
 FLOOR,COMMERCIAL BUILDING-1,KOHINOOR MALL,MUMBAI-70

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Investigation**Observed Value****Unit****Biological Reference Interval****Glucose Fasting**

(Plasma-F, Hexokinase)

124

mg/dL

Normal: 70-99

Impaired Fasting Glucose(IFG): 100-125

Diabetes mellitus: ≥ 126

(on more than one occasion)

(American diabetes association guidelines 2022)

Note:

An individual may show higher fasting glucose level in comparison to post prandial glucose level due to following reasons :

The glycaemic index and response to food consumed, Changes in body composition, Increased insulin response and sensitivity, Alimentary hypoglycemia, Renal glycosuria, Effect of oral hypoglycaemics & Insulin treatment.

Associated Tests:

HbA1c (H0018), Diabetes Profile – Maxi (D0021),HOMA Index (H0275), Insulin (I0275).

**Lipid Profile - 2 (Mini - Fasting)**

(Serum)

Cholesterol Total, Serum

(Enzymatic colorimetric Assay)

142

mg/dL

Desirable: < 200

Borderline High: 200-240

High: ≥ 240 **Triglycerides, Serum**

(Enzymatic)

148

mg/dL

Normal: < 150

Borderline High: 150-199

High: 200-499

Very High: ≥ 500 **HDL Cholesterol Direct**

(Homogeneous Enzymatic colorimetric Assay)

36

mg/dL

> 40

Non HDL Cholesterol

(Calculated)

106.5

mg/dL

Optimal: < 130

Desirable: 130-159

Borderline high: 159-189

High: 189-220

Very High: ≥ 220 **LDL Cholesterol**

(Calculated)

76.9

mg/dL

Optimal: < 100

Near Optimal: 100-129

Borderline high: 130-159

High: 160-189

Very High: ≥ 190 **VLDL Cholesterol**

(Calculated)

29.60

mg/dL

6-38



Alap
 Dr. ALAP CHRISTY

MD (Biochemistry)
 HOD - Clinical Chemistry
 Reg No.2020126991

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Investigation	Observed Value	Unit	Biological Reference Interval
LDL/HDL Ratio (Calculated)	2.2		2.5-3.5
Cholesterol / HDL Ratio (Calculated)	4.00		3.5-5

Note:

Reference Interval as per National Cholesterol Education Program (NCEP) Adult Treatment Panel III Report.

VLDL,CHOL/HDL RATIO,LDL/HDL RATIO,LDL Cholesterol,serum,Non HDL Colesterol are calculated parameters

**Bilirubin Total, Direct, Indirect**

(Serum)

Bilirubin Total (Diazo method)	1.00	mg/dL	0-1.2
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Interpretation :

1. Total Bilirubin is the sum of the unconjugated and conjugated fractions. Total Bilirubin is elevated in hepatitis, cirrhosis, haemolytic disorders , several inherited enzyme deficiencies, and conditions causing hepatic obstruction.
2. Neonatal Bilirubin quantitation is used to monitor diseases causing jaundice in the new-born, chiefly erythroblastosis fetalis (also caused haemolytic disease of the newborn or HDN.)
3. Physiologic jaundice is seen at serum bilirubin concentrations from 7 to 17 mg/dl. Serum bilirubin concentrations greater than 17 mg/dl may be pathologic. The primary concern is the potential for bilirubin encephalopathy or kernicterus.

Bilirubin Direct (Diazo method)	0.36	mg/dL	0.0-0.3
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Note: Direct Bilirubin is elevated in conditions causing hepatic obstruction , hepatitis, cirrhosis, several inherited enzyme deficiencies, and inherited defects in canalicular excretion.

Bilirubin- Indirect (Calculated)	0.64	mg/dL	0.1-1.0
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SGOT (AST) (Serum, IFCC w/o pyridoxal phosphate activation)	15	U/L	0-40
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SGPT (ALT) (Serum, IFCC w/o pyridoxal phosphate activation)	12	U/L	0-41
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Alkaline Phosphatase, Serum (Serum, p-nitrophenyl phosphate (pNPP) - (IFCC))	78	U/L	40-129
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Gamma GT (GGTP) (Serum, Enzymatic IFCC)	17	U/L	< 60
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Proteins, Serum (Serum)			
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Total Protein (Biuret test)	7.41	gm/dL	6.4-8.3
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Albumin, Serum (Bromocresol Green (BCG))	4.51	gm/dL	3.5-5.2
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Globulin	2.90	gm/dL	1.8-3.6
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Albumin/Globulin Ratio (Calculated)	1.6		1.1-2.2
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Dr. Alap Christy
Dr. ALAP CHRISTY
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 Reg No.2020126991

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Investigation	Observed Value	Unit	Biological Reference Interval
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Interpretation:

- Total Proteins are useful in the diagnosis and treatment of disease involving liver, kidney, bone marrow ,metabolic and nutritional disorders.
- The protein concentration of serum is an indicator of the hydration state of the body.
- Prolonged bed rest results in decreased total protein concentration.
- The A/G ratio measures the relative ratio of albumin to globulin
- Low A/G ratio may indicate viral infections, liver and kidney disease, or autoimmune disorders. These diseases increase globulin and decrease albumin thus lowering the A/G ratio.
- A high A/G ratio may indicate diseases that make the body produce less globulin, such as genetic disorders or may result from the use of immunosuppressive drugs.

Reference:

- Juraschek SP, Moliterno AR, Checkley W, Miller ER 3rd. The Gamma Gap and All-Cause Mortality. PLoS One. 2015 Dec 2;10(12):e0143494
- Busher JT. Serum Albumin and Globulin. In: Walker HK, Hall WD, Hurst JW, editors. Clinical Methods: The History, Physical, and Laboratory Examinations. 3rd edition. Boston: Butterworths; 1990. Chapter 101.
- Pack Insert

**LDH (Lactate Dehydrogenase), Serum** 175

U/L

0-250

(Serum, Lactate-pyruvate IFCC)

Interpretation:

- Lactate dehydrogenase (LDH) is an enzyme that is present in heart, liver, muscle and kidney tissues.
- Very high levels of LDH are seen in patients with megaloblastic anaemia, carcinoma and shock.
- Moderate increase seen in muscular disorders, nephrotic syndrome, and cirrhosis.
- Mild increases in LDH activity are seen in cases of myocardial or pulmonary infarction, leukemia, hemolytic anemia and non-viral hepatitis
- Many cancers cause a general increase in LDH levels or an increase in one of its isozymes.

Clinical Utility:

- It is used to screen for tissue damage along with other related tests like LDH isoenzymes.
- It is also used to monitor severe infections or conditions like hemolytic or megaloblastic anemias, kidney disease, and liver disease

Associated Test :

CPK Total Serum (C0165), SGPT ALT Serum (S0019), Troponin-I, Serum(T0128), Troponin-T, Serum (T0129) ,

LDH Isoenzymes (L0016)

Reference :

- Package Insert
- Klein R, Nagy O, Tóthová C, Chovanová F. Clinical and Diagnostic Significance of Lactate Dehydrogenase and Its Isoenzymes in Animals. Vet Med Int. 2020 Jun 15;2020:5346483.

**BUN, Serum**

9.6

mg/dL

6-20

(Serum, Urease)

Remark: In blood, Urea is usually reported as BUN and expressed in mg/dl. BUN mass units can be converted to urea mass units by multiplying by 2.14.

GFR with Creatinine (Above 18 Years)

(Serum)



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Investigation**Observed Value****Unit****Biological Reference Interval****Creatinine, Serum**

1.03

mg/dL

0.67-1.17

(Enzymatic (IDMS Traceable))

Kindly note change in Method and reference ranges

Interpretation

- Creatinine is a waste product formed in muscles from the high energy storage compound, creatine phosphate. The amount of creatinine produced is constant (unlike Urea) and is primarily a function of muscle mass.

Physiological factors affecting serum creatinine concentration includes age, gender, race, muscularity, exercise, Pregnancy, certain drugs, diet, dehydration and nutritional status.

Low serum Creatinine levels is seen in cases of low muscle mass like muscular atrophy, or aging.

High serum creatinine levels is seen in Acute and Chronic kidney disease, obstruction.

Since a rise in blood creatinine is observed only with marked damage of the nephrons, it is not suited to detect early stage kidney disease.

eGFR

88

ml/min/1.73

Normal OR high: ≥ 90

(Calculated)

sq m

Mild decrease : 60-89

Mild moderate decrease: 45-59

Moderate to severe decrease: 30-44

Severe decrease: 15-29

Kidney failure: < 15 **Interpretation:**

Kidneys filter the blood by removing waste and extra water to make urine. The glomerular filtration rate (GFR) shows how well the kidneys are filtering. When found early, important steps can be taken to protect their kidneys.

In adults, the normal eGFR number is usually more than 90. eGFR declines with age, even in people without kidney disease. See chart below for average estimated eGFR based on age.

Age (years)	Average eGFR
20-29	116
30-39	107
40-49	99
50-59	93
60-69	85
70+	75

Note:

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Investigation

Observed Value

Unit

Biological Reference Interval

- Race-'OTHERS' means other than black (Caucasian / Asians)
- Creatinine test done by IDMS traceable kit
- Patient flyer https://www.kidney.org/sites/default/files/01-10-8374_2210_patflyer_egfr.pdf
- Calculated by CKD-EPI Creatinine Equation 2021.



Uric Acid, Serum

(Serum, Uricase)

6.8

mg/dL

3.4-7.0

Interpretation:

- Increased in Gout, asymptomatic hyperuricemia, leukemia, polycythemia, hemolytic anemia, sickle cell anemia, resolving pneumonia, toxemia of pregnancy, psoriasis, lymphoma, metabolic acidosis, chronic lead poisoning.
- Decreased in disorders of copper accumulation, kidney tubule disorder, Acromegaly, Celiac disease, Xanthine oxidase deficiency.
- Its used to monitor gout and also chemotherapeutic treatment of neoplasm to avoid renal urate deposition with possible renal failure (tumor lysis syndrome).

Note:

- A purine rich diet as well as sever exercise increases uric acid values.
- High protein-weight reduction diet and alcohol consumption can cause raised uric acid levels.

Reeference:

- Package insert
- Wallach's interpretation of diagnostic tests, Ed11, 2020.
- Henry's Clinical Diagnosis and Management by Laboratory Methods. 23rd ed; 2017.
- Tietz fundamentals of clinical chemistry 6th edition. Burtis CA, Ashwood ER, Bruns DE, 2008.

Electrolytes, Serum

(Serum, ISE Indirect)



Sodium, Serum

138

mmol/L

136-145

Interpretation:

- Low levels are noted in prolonged vomiting or diarrhea, diminished reabsorption in the kidney and excessive fluid retention. High levels are seen in case of excessive fluid loss, high salt intake and increased kidney reabsorption



Potassium, Serum

4.80

mmol/L

3.5-5.1

Interpretation:

- Low levels are noted in reduced intake of dietary potassium or excessive loss of potassium from the body due to diarrhea, prolonged vomiting or increased renal excretion. High levels may be caused by dehydration or shock, severe burns, hemolysis, diabetic ketoacidosis, and retention of potassium by the kidney



Chloride, Serum

100

mmol/L

98-107

Interpretation:

- **Low levels** are noted in reduced dietary intake, prolonged vomiting and reduced renal reabsorption as well as some forms of acidosis and alkalosis. High levels are found in dehydration, kidney failure, some forms of acidosis, high dietary or parenteral chloride intake, and salicylate poisoning.



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Investigation	Observed Value	Unit	Biological Reference Interval
ESR - Erythrocyte Sedimentation Rate,EDTA (EDTA Whole Blood) Method : Manual	6	mm/hr	0-15

Interpretation :

1. It indicates presence and intensity of an inflammatory process, never diagnostic of a specific disease. Changes are more significant than a single abnormal test.
2. It is a prognostic test and used to monitor the course or response to treatment of diseases like tuberculosis, bacterial endocarditis, acute rheumatic fever, rheumatoid arthritis, SLE, Hodgkins disease, temporal arteritis, polymyalgia rheumatica.
3. It is also increased in pregnancy, multiple myeloma, menstruation, and hypothyroidism.

**HbA1c- Glycated Haemoglobin**

(EDTA Whole Blood)

HbA1c- Glycated Haemoglobin 5.8

(High-Performance Liquid Chromatography (HPLC))

%

Non-diabetic: <= 5.6

Pre-diabetic: 5.7-6.4

Diabetic: >= 6.5

Refer interpretation for monitoring ranges.

Estimated Average Glucose (eAG) 119.76 mg/dL
 (Calculated)

Interpretation & Remark:

1. HbA1c is used for monitoring diabetic control. It reflects the estimated average glucose (eAG).
2. HbA1c has been endorsed by clinical groups & ADA (American Diabetes Association) guidelines 2017, for diagnosis of diabetes using a cut-off point of 6.5%.
3. Trends in HbA1c are a better indicator of diabetic control than a solitary test.
4. Low glycated haemoglobin(below 4%) in a non-diabetic individual are often associated with systemic inflammatory diseases, chronic anaemia(especially severe iron deficiency & haemolytic), chronic renal failure and liver diseases. Clinical correlation suggested.
5. To estimate the eAG from the HbA1c value, the following equation is used: $eAG(mg/dl) = 28.7 * A1c - 46.7$
6. Interference of Haemoglobinopathies in HbA1c estimation.

- A. For HbF > 25%, an alternate platform (Fructosamine) is recommended for testing of HbA1c.
- B. Homozygous hemoglobinopathy is detected, fructosamine is recommended for monitoring diabetic status
- C. Heterozygous state detected (D10/ Tosho G8 is corrected for HbS and HbC trait).



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 HOD Hematology & Transplant Immunology
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Investigation**Observed Value****Unit****Biological Reference Interval**

7. In known diabetic patients, following values can be considered as a tool for monitoring the glycemic control.

Excellent Control - 6 to 7 %,
Fair to Good Control - 7 to 8 %,
Unsatisfactory Control - 8 to 10 %
and Poor Control - More than 10 % .

Note : Hemoglobin electrophoresis (HPLC method) is recommended for detecting hemoglobinopathy.



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HOD Hematology & Transplant Immunology
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Routine Examination Profile - Urine

Investigation	Observed Value	Unit	Biological Reference Interval
Routine Examination Profile - Urine			
GENERAL EXAMINATION			
Colour	Yellow		Pale Yellow
Transparency (Appearance)	Clear		Clear
Reaction (pH)	5.5		4.5-8
Specific Gravity	1.011		1.005-1.025
CHEMICAL EXAMINATION (AUTOMATED URINE CHEMISTRY)			
Urine Protein (Albumin)	Absent		Absent
(Protein Error Principle)			
Urine Ketones (Acetone)	Absent		Absent
(Legals Test)			
Urine Glucose (Sugar)	Absent		Absent
(Glucose Oxidase-Peroxidase (GOD-POD))			
Urobilinogen	Normal		Normal
(Diazo method)			
Bilirubin	Negative		Negative
(Diazo method)			
Nitrite	Negative		Negative
(Greiss Reaction)			
MICROSCOPIC EXAMINATION(CUVETTE BASED IMAGING TECHNOLOGY)			
Red blood cells	0.0	/hpf	0-2
Dysmorphic Red Blood Cells	Absent		Absent
Pus cells (WBCs)	0.9	/hpf	0-5
Epithelial cells	0.3	/hpf	0-5
Crystals	0.2	/hpf	0-1.36
CRY - Calcium-oxalate monohydrate	0.0		0-1.36
CRY - Calcium-oxalate dihydrate	0.0		0-1.36
CRY - Triple-phosphate	0.0		0-1.36
CRY - Uric acid	0.0		0-1.36
Casts - Hyalin	0.0	/hpf	0-2
Casts - Pathological	0.0	/hpf	0-0.34



Niranjan Patil

Dr. NIRANJAN PATIL
MD (Microbiology)
HOD - Microbiology & Molecular Biology
Reg No.2006020697

MEDICAL LABORATORY REPORT

Registered On: 09/09/2025 9:20 AM

Collected On: 09/09/2025 9:08 AM

Reported On: 09/09/2025 10:00 PM

PROCESSING LOCATION :

METROPOLIS HEALTHCARE LTD,UNIT NO409-416,4TH
FLOOR,COMMERCIAL BUILDING-1,KOHINOOR MALL,MUMBAI-70

** Referred Test

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Name : Mr. SANTOSH VENKITACHALAM
 Age / Gender : 50 Year(s)/ Male
 Contact No. : +919920105605
 Address : A 502 MULUND KUMAR MHADA COLONY M...
 Pin code : 400081

VID No. : 250067508503480
 PID No. : P18825553373078
 Referred by : DR.ABHJEET JADHAV
 Sample Collected At : Hcl Avitas Private Limited, 806,siddharth 96 Nehru Place,110019

Investigation	Observed Value	Unit	Biological Reference Interval
Bacteria	27.6	/hpf	0-65.00
Trichomonas Vaginalis	Absent		Absent
Yeast cells	0.0	/hpf	0-0.68

Test Interpretation:

1. Urine Routine and Microscopy is a screening test designed to detect early signs of renal, metabolic, and infectious diseases.
2. Abnormal results of chemical examination are confirmed by manual methods.
3. The automated urine analyser uses **AI-assisted** image analysis for urinary sediment evaluation. The system enhances the consistency, objectivity, and accuracy of microscopic reporting by scanning multiple fields and applying trained algorithms to classify and quantify particles.

4. Chemical examination is performed using dipstick methods, with the following principles

- Protein – Protein Error Principle
- Glucose – Glucose Oxidase-Peroxidase Method
- Ketone – Legal's Test
- Bilirubin – Azo-Diazo Reaction
- Urobilinogen – Diazonium Ion Reaction
- Nitrite – Griess Method

1. AI interpretation of microscopy includes the classification of urine elements such as RBCs, WBCs, epithelial cells, bacteria, crystals, casts, and others, with flagging of abnormalities. AI results are reviewed and validated by qualified personnel.

2. All abnormal results from chemical examination are confirmed using manual methods, including:

- **pH** – pH paper
- **Specific Gravity** – Urinometer
- **Protein** – Sulfosalicylic Acid Method
- **Glucose** – Benedict's Method
- **Ketone** – Rothera's Method
- **Bile Salt** – Sulfur Granule Method
- **Bile Pigment** – Fouchet Method
- **Urobilinogen** – Ehrlich's Method
- **Nitrite** – Nitrate Reduction Test

1. Pre-test sample collection recommendations:

- First-void, mid-stream urine collected in a clean, dry, sterile container
- Avoid contamination from vaginal, urethral, or perineal discharges
- Avoid prolonged transit and exposure to sunlight, which may alter test results

2. A **negative nitrite** test does not rule out bacterial infection or Urinary Tract infection.



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Investigation**Observed Value****Unit****Biological Reference Interval**

3. **Trace proteinuria** can occur under physiological conditions such as prolonged recumbency, exercise, or high protein intake.
4. **False-positive/negative reactions** for bile pigments, proteins, glucose, and nitrites may occur due to interfering substances such as disinfectants, dyes, **ascorbic acid**, or certain medications.
5. **Physiological variations** (hydration status, diet, physical activity, etc.) may influence urine composition.
6. **AI-assisted microscopy findings** are reported in **decimal values**, reflecting the **arithmetic mean** of particles identified across multiple scanned fields, improving quantitative accuracy.

Disclaimer:

1. Images will not be available for normal (results within the normal range) microscopy results/findings.
2. A careful consideration to combination of epidemiological factors, stage of infection, clinical history, examination, other relevant investigation findings and treatment history should be done when interpreting test results.

Associated Test:

1. Culture & sensitivity, Urine (Test code: C0202)

Clinical references:

- Akin OK, Serdar MA, Cizmeci Z, Genc O, Aydin S. Comparison of LabUMat-with-UriSed and iQ200 fully automatic urine sediment analysers with manual urine analysis. Biotechnol Appl Biochem. 2009 Jun;53 (Pt 2) :139-44. doi: 10.1042/BA20080188. PMID: 19021504.

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Investigation**Observed Value****Unit****Biological Reference Interval**

Calcium, Serum
 (Serum, NM-BAPTA)

9.3

mg/dL

8.6-10.0

Iron Studies

(Serum)

Iron, Serum
 (FerroZine)

128.00

µg/dL

33-193

UIBC

(FerroZine)

190.00

µg/dL

125-345

TIBC

(Calculated)

318

µg/dL

250-450

Transferrin Saturation

(Calculated)

40.25

%

14-50

Interpretation :

1. Measurements of serum iron, TIBC and the percentage of iron saturation of transferrin are useful screening tests for iron deficiency anaemia.
2. However, serum iron exhibits significant diurnal variation and may transiently rise or reach reference values after dietary or iron supplements & post blood transfusion.
3. The diagnostic specificity of a low serum iron for iron deficiency is lost in the presence of acute & chronic inflammatory processes as the concentrations of iron and transferrin in the serum are significantly affected, and fall rapidly as part of the acute phase response irrespective of the iron stores status in the body.
4. Hence, Concurrent measurement of the markers mentioned in the below interpretative table alongwith serum iron studies improves the diagnostic specificity for iron deficiency anaemia & also provides a reliable work up for microcytic hypochromic anaemia.

Tests	Iron Deficiency anaemia	Anaemia of Chronic disease	Iron overload	Hemoglobinopathy (Especially Trait)
Serum Iron	Decreased	Decreased	Increased	Normal
Serum Total Iron Binding Capacity	Increased	Decreased or Normal	Increased or Normal	Normal
% Transferrin Saturation	Decreased	Decreased or Normal	Increased or Normal	Normal
Serum UIBC	Increased	Decreased or Normal	Decreased	Normal
Serum Ferritin	Decreased	Increased	Increased or Normal	Normal
Serum Soluble Transferrin receptor	Increased	Normal	Decreased	Normal
Serum Hepcidin	Normal	Increased	Normal	Normal

Associated Tests :

1. Serum Soluble Transferrin receptor
2. Serum Hepcidin



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 Consultant Biochemist
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Investigation**Observed Value****Unit****Biological Reference Interval****Interpretation :**

Measurements of serum iron, UIBC-Unsaturated Iron binding Capacity and the percentage of iron saturation of transferrin are useful screening tests for iron deficiency anemia.

Tests	Iron Deficiency anemia	Anemia of Chronic disease	Iron overload	Hemoglobinopathy (Especially Trait)
Serum Iron	Decreased	Decreased	Increased	Normal
Serum Total Iron Binding Capacity	Increased	Decreased or Normal	Increased or Normal	Normal
% Transferrin Saturation	Decreased	Decreased or Normal	Increased or Normal	Normal
Serum UIBC	Increased	Decreased or Normal	Decreased	Normal
Serum Ferritin	Decreased	Increased	Increased or Normal	Normal
Serum Soluble Transferrin receptor	Increased	Normal	Decreased	Normal
Serum Hepcidin	Normal	Increased	Normal	Normal

- The sum of the serum iron and UIBC represents total iron-binding capacity (TIBC) and is a measurement for the maximum iron concentration that the iron complex can bind.

Clinical Utility:

- Iron measurements are used in the diagnosis of iron deficiency anemia, hemochromatosis, microcytic anemia, macrocytic anemia, erythropoietin deficiency. The test is also undertaken in patients of chronic renal disease, hemolytic anemia and hemoglobinopathy.

Caution:

- Serum iron exhibits significant diurnal variation and may transiently rise or reach reference values after dietary or iron supplements & post blood transfusion.
- Patients treated with Deferoxamine containing drugs may give low values with the UIBC assay.

Associated Tests:

Soluble Transferrin Receptor Serum (S0036), Serum Hepcidin (H0077).Serum Ferritin (F0018)

Reference:

- Package insert
- Wallach's interpretation of diagnostic tests, Ed11, 2020
- Henry's Clinical Diagnosis and Management by Laboratory Methods. 23rd ed; 2017.
- Tietz fundamentals of clinical chemistry 6th edition. Burtis CA, Ashwood ER, Bruns DE, 2008.

**Phosphorus, Serum**

(Serum, Molybdate UV)

4.1

mg/dL

2.5-4.5

**RA (Rheumatoid Arthritis) Factor, Serum**

(Serum, Immunoturbidimetry)

Below 10

IU/mL

Non-reactive: < 14

Interpretation :

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Investigation**Observed Value****Unit****Biological Reference Interval**

- Increased in RA, Chronic hepatitis, chronic viral infections, cirrhosis, Infectious mononucleosis, leishmaniasis, leprosy, malaria sarcoidosis Syphilis, TB etc.
- High levels are also seen in inflammatory rheumatic diseases and in various non rheumatic diseases.

Clinical Utility:

- The detection of rheumatoid factor (RF) is one of the criteria of the American Rheumatism Association (ARA) for the diagnosis of Rheumatoid Arthritis (RA) especially when clinical diagnosis is difficult.
- RFs play an important role in the differential diagnosis between RA and other rheumatic diseases. They also permit prognostic statements with regards to RA.

Note:

- Values of RA may increase with age. It should be interpreted along with overall clinical picture.
- Recent blood transfusion, multiple vaccinations or transfusions may affect results.
- Serum with cryoglobulin or high lipid levels may cause false-positive test results.

Associated Tests:

Rheumatoid Arthritis Panel (R0022), Cyclic Citrullinated Peptide Antibody (CCP Antibody) Serum (C0047)

Reference:

- Package insert
- Wallach's interpretation of diagnostic tests, Ed11, 2020
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Investigation**Observed Value****Unit****Biological Reference Interval****Thyroid Profile - 1**

(Serum, Electrochemiluminescence immunoassay (ECLIA))

T3 (Total Triiodothyronine) 136.00 ng/dL 84.6-201.8

T4 (Total Thyroxine) 11.10 µg/dL 5.1-14.1

TSH (Thyroid Stimulating Hormone) - 2.670 µIU/mL 0.54-5.3

Ultrasensitive, Serum

INTERPRETATION

TSH	T3 / FT3	T4 / FT4	Suggested Interpretation for the Thyroid Function Tests Pattern
Within Range	Decreased	within Range	•Isolated Low T3-often seen in elderly & associated Non-Thyroidal illness. In elderly the drop in T3 level can be upto 25%.
Raised	Within Range	Within Range	•Isolated High TSH especially in the range of 4.7 to 15 mIU/ml is commonly associated with Physiological & Biological TSH Variability. •Subclinical Autoimmune Hypothyroidism •Intermittent T4 therapy for hypothyroidism •Recovery phase after Non-Thyroidal illness
Raised	Decreased	Decreased	•Chronic Autoimmune Thyroiditis •Post thyroidectomy, Post radioiodine •Hypothyroid phase of transient thyroiditis
Raised or within Range	Raised	Raised or within Range	•Interfering antibodies to thyroid hormones (anti-TPO antibodies) •Intermittent T4 therapy or T4 overdose •Drug interference- Amiodarone, Heparin, Beta blockers, steroids, anti-epileptics
Decreased	Raised or within Range	Raised or within Range	•Isolated Low TSH -especially in the range of 0.1 to 0.4 often seen in elderly & associated with Non-Thyroidal illness •Subclinical Hyperthyroidism •Thyroxine ingestion
Decreased	Decreased	Decreased	•Central Hypothyroidism •Non-Thyroidal illness •Recent treatment for Hyperthyroidism (TSH remains suppressed)
Decreased	Raised	Raised	•Primary Hyperthyroidism (Graves' disease), Multinodular goitre, Toxic nodule •Transient thyroiditis: Postpartum, Silent (lymphocytic), Postviral (granulomatous, subacute, DeQuervain's), Gestational thyrotoxicosis with hyperemesis gravidarum
Decreased or within Range	Raised	Within Range	•T3 toxicosis •Non-Thyroidal illness

References: 1. Interpretation of thyroid function tests. Dayan et al. THE LANCET • Vol 357 • February 24, 2001

2. Laboratory Evaluation of Thyroid Function, Indian Thyroid Guidelines, JAPI, January 2011, vol. 59

Vitamin D Total - 25 Hydroxy (OH)**26.10**

ng/mL

Insufficiency: 20-30

(Serum, Electrochemiluminescence immunoassay (ECLIA))

Sufficiency: 30-100

Hypervitaminosis: > 100

Deficiency: < 20

Vitamin B12 (Cyanocobalamin)**266.00**

pg/mL

197-771

(Serum, Electrochemiluminescence immunoassay (ECLIA))

-- End of Report --

Test Marked with NABL symbol are in the scope of accreditation



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